

Power BI Implementation & DAX Guide

This reference document details the step-by-step ETL transformations, star schema relationship bindings, and standard DAX measures used to build the AI Recruitment and Workforce Analytics Dashboard.

1. Data Model & Star Schema Relationships

The dashboard is structured as a STAR SCHEMA. Ensure you set up the following relationships in Power BI Model View:

From Table (Fact)	From Column	To Table (Dimension)	To Column	Cardinality	Direction
Fact_Recruitment	Candidate_Key	Dim_Candidate	Candidate_Key	Many-to-One (*:1)	Single
Fact_Recruitment	Department_Key	Dim_Department	Department_Key	Many-to-One (*:1)	Single
Fact_Recruitment	Recruiter_Key	Dim_Recruiter	Recruiter_Key	Many-to-One (*:1)	Single
Fact_Recruitment	Application_Date_Key	Dim_Date	Date_Key	Many-to-One (*:1)	Single (Active)
Fact_Recruitment	Offer_Date_Key	Dim_Date	Date_Key	Many-to-One (*:1)	Single (Inactive)
Fact_Recruitment	Joining_Date_Key	Dim_Date	Date_Key	Many-to-One (*:1)	Single (Inactive)

2. Essential DAX Measures

Total Applications:

```
Total Applications = COUNTROWS(Fact_Recruitment)
```

Total Hired:

```
Total Hired = CALCULATE(COUNTROWS(Fact_Recruitment), Fact_Recruitment[Status] = "Hired")
```

Hiring Rate (%):

```
Hiring Rate = DIVIDE([Total Hired], [Total Applications], 0)
```

Acceptance Rate (%):

```
Acceptance Rate =  
VAR Offered = CALCULATE(COUNTROWS(Fact_Recruitment), Fact_Recruitment[Status] IN {"Offered"  
    , "Hired"})  
RETURN DIVIDE([Total Hired], Offered, 0)
```

Avg Hiring Days:

```
Avg Hiring Days =  
AVERAGEX(  
    FILTER(Fact_Recruitment, NOT ISBLANK(Fact_Recruitment[Offer_Date_Key])),  
    RELATED(Dim_Date[Full_Date]) - RELATED(Dim_Date[Full_Date]) -- Link Date-to-Date  
)
```

Avg ATS Score:

```
Avg ATS Score = AVERAGE(Fact_Recruitment[Resume_ATS_Score])
```

Avg Salary Offered:

```
Avg Salary = AVERAGE(Fact_Recruitment[Offered_Salary])
```

Offer Conversion (%):

```
Offer Conversion = DIVIDE(CALCULATE(COUNTROWS(Fact_Recruitment), Fact_Recruitment[Status] =  
    "Offered"), [Total Applications], 0)
```